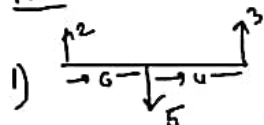


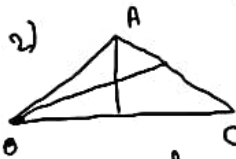
Geo

MPC



$$m_1 + m_2 = M$$

$$m_1 \times d_1 = m_2 \times d_2$$

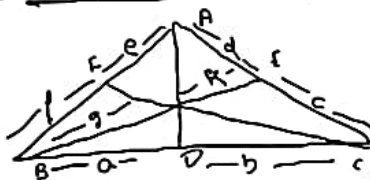


Cevian line $\rightarrow$ No mass distribution

Transverse Mass dist $\perp$



3) Ceva's Theorem P1



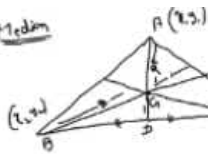
$$\frac{a}{b} \times \frac{c}{d} \times \frac{e}{f} = 1$$

Ceva's Theorem P2

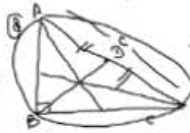
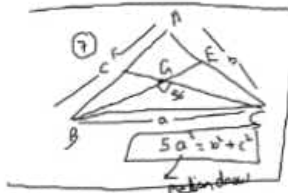
$$\frac{a}{b} = \frac{1}{e} + \frac{a}{b}$$

Use of MPC $\leftrightarrow$ Ratio $\leftrightarrow$

Part 2 Median

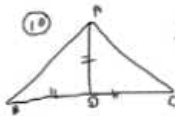


- 1) $BD = CD$
- 2) $Area(\triangle ABC) = Area(\triangle BDC)$
- 3) 1 median = 2 equal A
- 4) 3 medians = 6 equal A
- 5) $\frac{AG}{GD} = \frac{2}{1}$
- 6) $G \rightarrow \frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}$

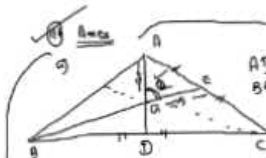


$BD = \frac{1}{2} BC$
 Circle radius
 $5m^2 = m^2 + m^2$

9) For any $\triangle \rightarrow$ Sum of medians $\rightarrow$ longer side
 Sum of medians $\rightarrow$ shorter side

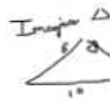


Sum of medians
 Apollonius
 $1) AB^2 + AC^2 = 4(AD^2 + \left(\frac{BC}{2}\right)^2)$
 $2) 3x = 4y \rightarrow 3(\text{side sum}) = 4(\text{median sum})$



$AD = 12$
 $BE = 15$
 $\frac{AG}{GD} = \frac{2}{1}$
 $AG = \frac{2}{3} \times 12 = 8$
 $\frac{BG}{GD} = \frac{2}{1}$
 $BG = \frac{2}{3} \times 15 = 10$
 $Area(\triangle ABC) = \frac{1}{2} \times 12 \times 10 \times \sin \theta = 60 \sin \theta$
 $Area(\triangle ABC) = 6 \times 12 \times 10$

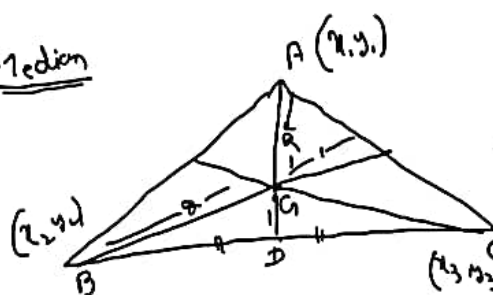
b) Median of 3 medians



$ABC = 16$
 $AB = 12$
 $AC = 10$

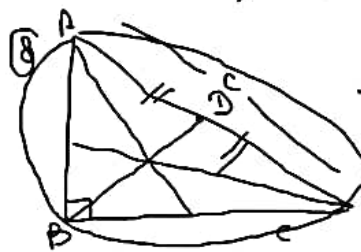
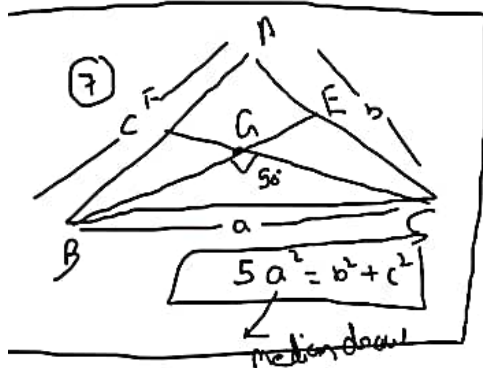


Part 2 Median



- 1) $BD = CD$
- 2) $\text{Area}(\triangle ABD) = \text{Area}(\triangle ADC)$
- 3) 1 median = 2 equal A
- 4) 3 median = 6 equal A
- 5) $\frac{AG}{GD} = \frac{2}{1}$

$$G \rightarrow \frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}$$



$$BD = \frac{4R^2 - a^2}{2}$$

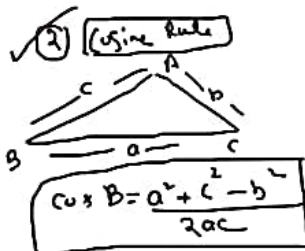
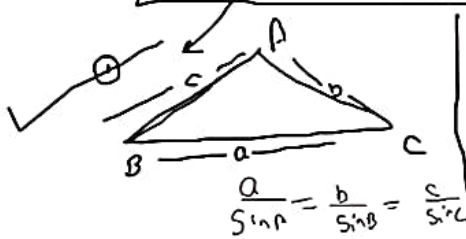
Circum Radius

$$5m_b^2 = ma^2 + mc^2$$

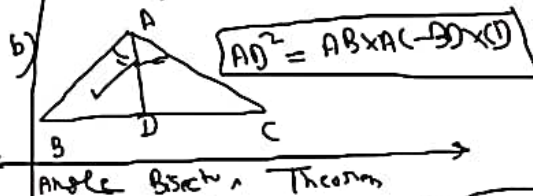
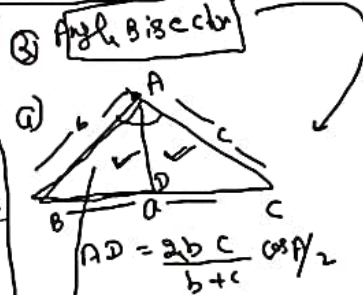
Q) For any $\triangle \rightarrow$ Shortest median $\rightarrow$ longest side
 longest median $\rightarrow$ Shortest side



Part 3 Sin, Cos, Side Rule, Cosine, Angle Bisector



If 2 side & 1 angle is known then find 3rd side.

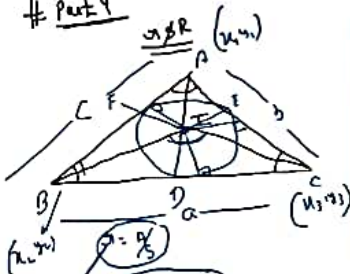


Angle Bisector Theorem

$$\frac{AB}{AC} = \frac{BD}{CD} \iff \frac{AB}{BD} = \frac{AC}{CD}$$



Part 4



$$r = \frac{A}{s}$$

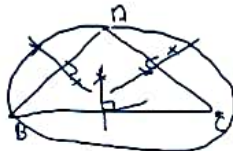
$$r = \frac{a}{2\sin A} \text{ (Eqn)}$$

$$r = \frac{a+b-c}{2}$$

$$\angle BIC = 90^\circ + \frac{\angle BAC}{2}$$

$$x = \frac{a^2 + b^2 + c^2}{2(a+b+c)}$$

$$y = \frac{a^2 + b^2 + c^2}{2(a+b+c)}$$



$$R = \frac{abc}{4A}$$

$$R = \frac{a}{2\sin A} \rightarrow \text{(equilateral)} \rightarrow R = \frac{a}{\sqrt{3}}$$

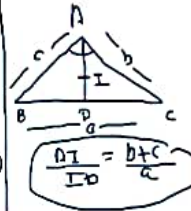
$$R = \frac{c}{2} \rightarrow \text{(right triangle)}$$

$$\text{Dis b/w Incenter \& Circumcenter}$$

$$D = \sqrt{R^2 - 2r^2}$$

$$\text{Area of Right angle tri if } \sqrt{R^2 - 2r^2}$$

$$\sqrt{R^2 - 2r^2}$$



$$\frac{OI}{IO} = \frac{b+c}{a}$$

all Polygon

Part 5



- ✓ 1) Obtuse $\rightarrow c^2 > a^2 + b^2$ (c is longest side)
- 2) Acute $\rightarrow c^2 < a^2 + b^2$ ()
- 3) Right $\rightarrow c^2 = a^2 + b^2$ ()



$\rightarrow$ Existence

Sum of two sides $>$ 3rd side
diff of two sides $<$ 3rd side

Point

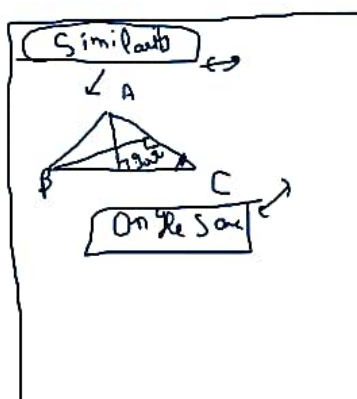
BPT



$$\frac{a}{b} = \frac{c}{d}$$

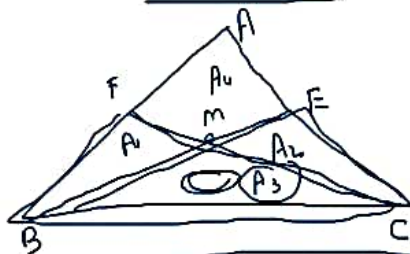


$$\frac{\Delta ABD}{\Delta ADC} = \frac{BD}{DC}$$

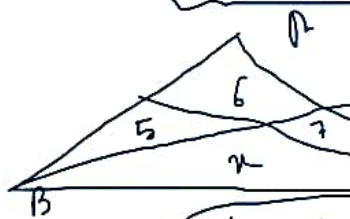


Part 7

Ladder theorem

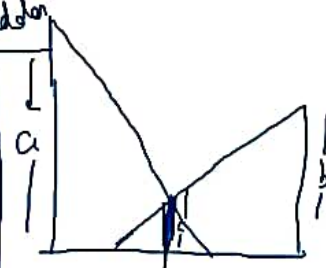


$$\frac{1}{A_1 + A_3} + \frac{1}{A_2 + A_3} = \frac{1}{A_3} + \frac{1}{A_1 + A_2 + A_3 + A_4}$$



$$\frac{1}{5+u} + \frac{1}{7+u} = \frac{1}{u} + \frac{1}{d+u}$$

Cross Ladder

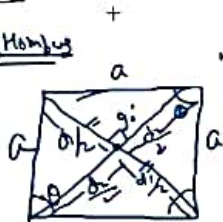


$$\frac{1}{c} = \frac{1}{a} + \frac{1}{b}$$



Part 8

Rhombus



$$1) 4a^2 = d_1^2 + d_2^2$$

$$2) A = \frac{1}{2} d_1 d_2$$

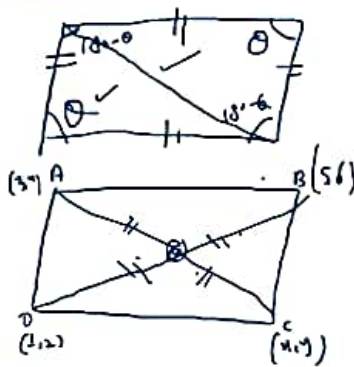
$$A = a^2 \sin \theta$$

$$3) \theta = \frac{\pi}{2}$$

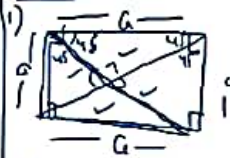
$$a = d_1 d_2$$

$$2 \sqrt{d_1^2 + d_2^2}$$

Parallelogram



Square



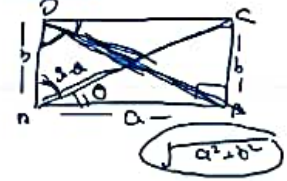
diagonals $\leftrightarrow a\sqrt{2}$

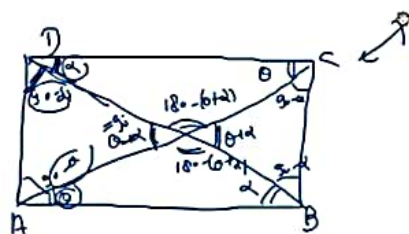


$$\text{Sum of all four Areas}$$

$$= 2a^2$$

③ Rectangle

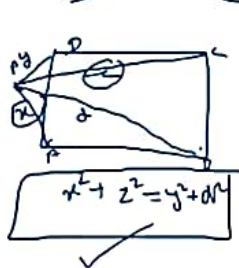




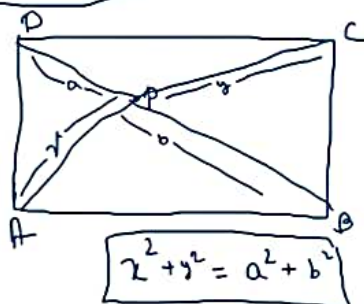
$$90 - d = a$$

$$90 - c = b$$

Theorem $\leftarrow$ British Flag Theorem

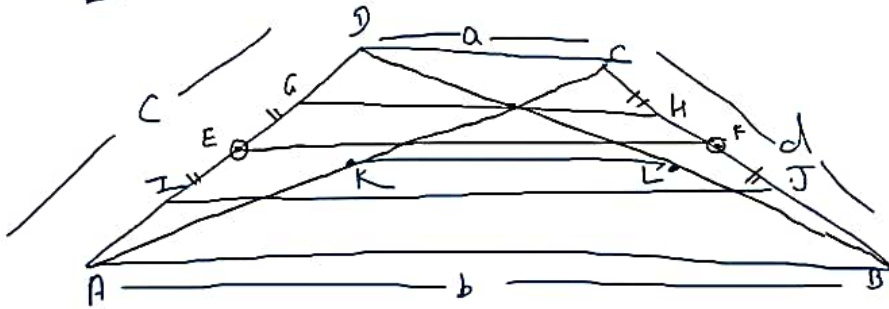


$$x^2 + z^2 = y^2 + d^2$$



$$x^2 + y^2 = a^2 + b^2$$

Part 9 → Trapezium

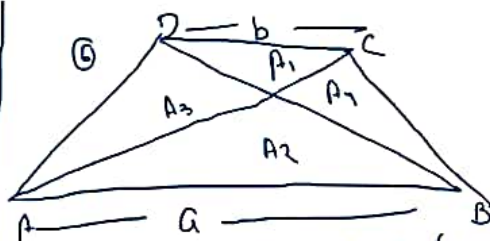
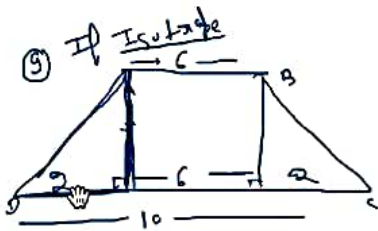


④ $KL = \frac{b-a}{2}$ (midpoint of diag) ① $EF = \frac{a+b}{2}$

② $GH = \frac{2ab}{a+b}$

⑤ $p^2 + q^2 = c^2 + d^2 + 2ab$ (length of diagonals) ③ I_T (Divide Area into 2 equal part)
 $I_T = \sqrt{\frac{c^2 + d^2}{2}}$

⑧ Area of trapezium
 $\frac{1}{2}(\text{Sum of Para sides}) \times H$

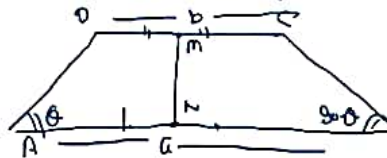


$A_1, A_2 = A_3, A_4$ (for all trapezium)

If $AB = CD$ → Trapezium

$A_1, A_2 = A_3, A_4$

⑦

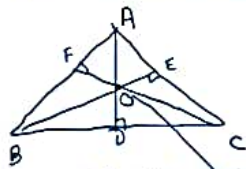


$m = \frac{a-b}{2}$

Restriction

Base angle sum = 360°

Part 10 Orthocentre Altitude



$$1) AO \times OD = BO \times OE = CO \times OF$$

$$2) \frac{AO}{AD} + \frac{BO}{BE} + \frac{CO}{CF} = 2$$

$$3) H_1, H_2, H_3 \text{ (altitudes)}$$

$$\frac{H_1 H_3}{H_1 + H_3} < H_2 < \frac{H_1 H_3}{H_1 - H_3}$$

$$4) 2, 5, 11$$

$$\frac{2 \times 5}{5 + 2} < H < \frac{2 \times 5}{5 - 2}$$

$$\frac{10}{7} < H < \frac{10}{3}$$

$$1.4 < H < 3.3$$

$$H = 2, 3$$

$$4) \text{Side 2} \perp \text{Alt}$$

$$\text{Side 2} \perp \text{height}$$

$$H_1, H_2, H_3$$

$$2:3:4$$

$$0:b:c$$

$$\frac{1}{2}:\frac{1}{3}:\frac{1}{4}$$

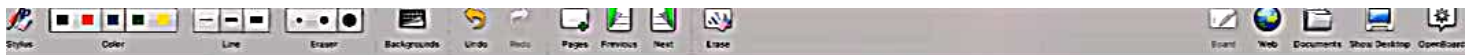
Right angle triangles Orthocentre
lie on go vertex

5) Viviani's Theorem

only for equilateral Δ



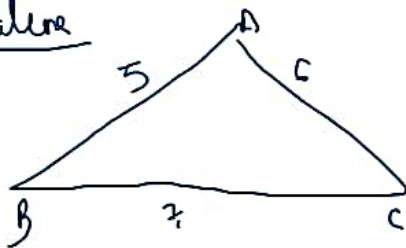
$$H_1 + H_2 + H_3 = H \text{ of equilateral}$$



Part 11

Diff type of triangle Area

Scalene

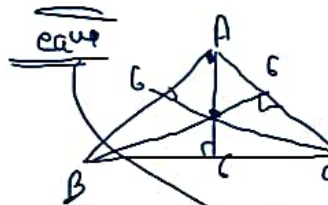
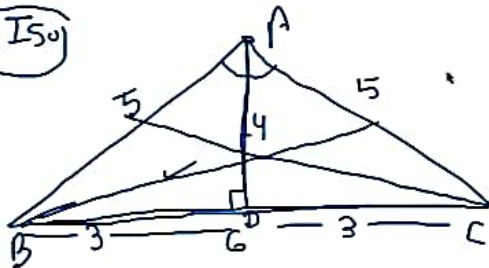


Apply Heron's

Ex Euler's



Iso



$$A = \frac{1}{2} \times 6 \times 4 = \frac{\sqrt{3}}{2} \times 6^2$$

$$A = \frac{9\sqrt{3}}{2}$$

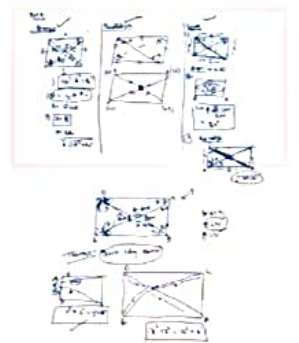
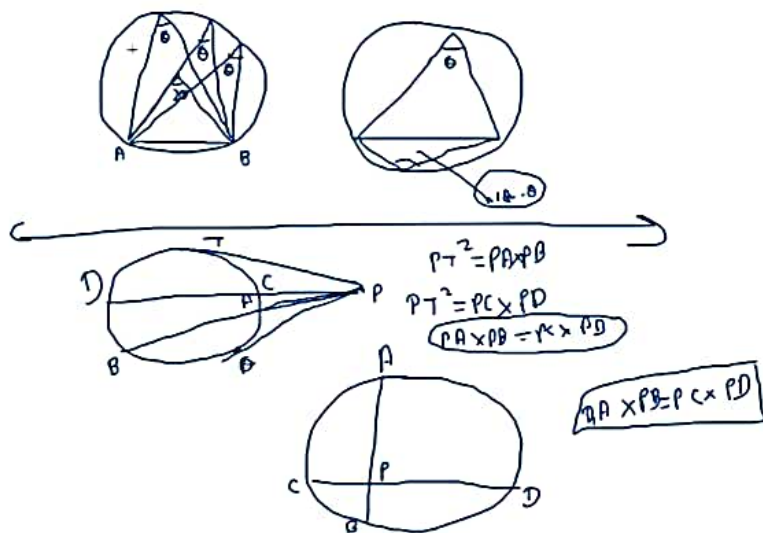
$$A = \frac{C\sqrt{3}}{2}$$

$$A = G \sin 60^\circ$$





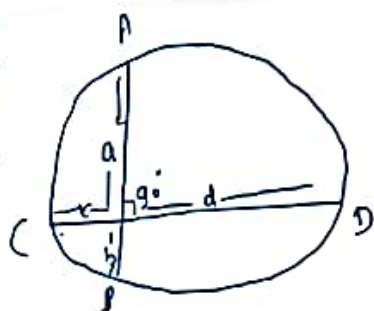
Part 2





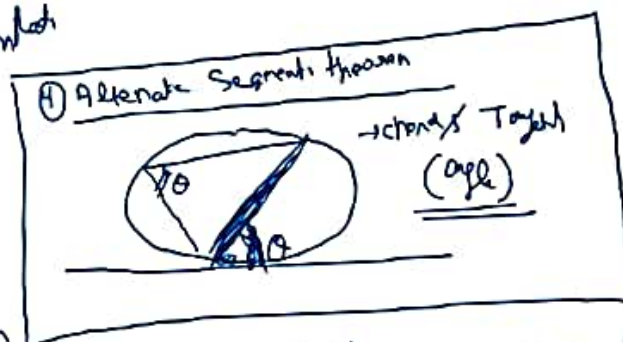
Part 13

1)



$$R^2 = \frac{a^2 + b^2 + c^2 + d^2}{4}$$

Circumradius



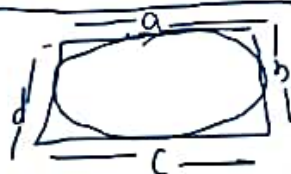
2)



Radius will be in GP

3)

for any Polygon of even sided



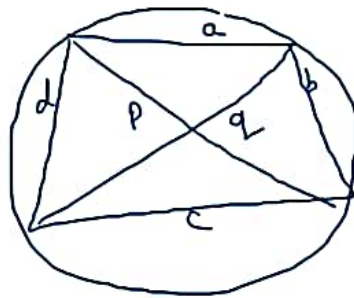
$$a - b + c - d = 0$$



$$a - b + c - d + e - f = 0$$



Part 14 $\longleftrightarrow$ Cyclic Quadrilateral
 opposite angle $\rightarrow 180^\circ$



Area of cyclic
 quadrilateral

$$\sqrt{(s-a)(s-b)(s-c)(s-d)}$$

Ptolemy's Theorem

$$p \times q = a \times c + b \times d$$

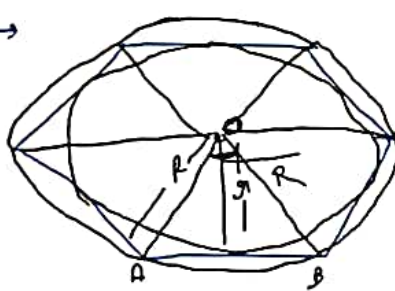


Part 15 $\leftrightarrow$ Polygon

1) Sum of Internal Angle $= (n-2) \pi$
Regular Polygon $\leftrightarrow$ Angle $\rightarrow \frac{(n-2)\pi}{n}$

2) Sum of External Ang $\rightarrow 360^\circ$

Regular Poly $\leftrightarrow$



Area of Polygon

$$\frac{n a^2 \cot\left(\frac{180^\circ}{n}\right)}{4}$$



$$\sin\left(\frac{180^\circ}{n}\right) = \frac{a}{2R}$$

$$\cos\left(\frac{180^\circ}{n}\right) = \frac{a}{R}$$

$$\tan\left(\frac{180^\circ}{n}\right) = \frac{a}{2a}$$

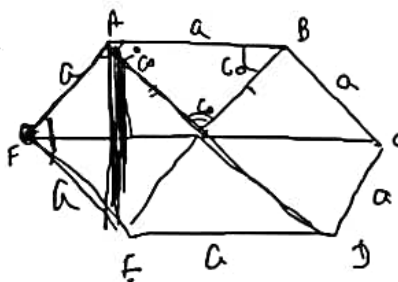
ΔOAB

$$\frac{1}{2} \times a \times a$$

$$\frac{1}{2} \times a \times \frac{a}{2} \cot\left(\frac{180^\circ}{n}\right)$$

$$\frac{a^2}{4} \cot\left(\frac{180^\circ}{n}\right)$$

Regular Hexagon



Area of Reg Polygon (Hexagon)

$$6 \times \frac{\sqrt{3}}{4} \times a^2$$

2 Diagonal

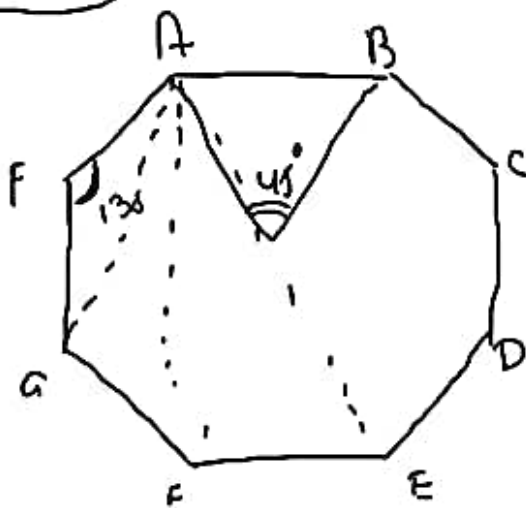


$$6 \rightarrow \sqrt{3} a$$

$$3 \rightarrow 2a$$

Part 16

Octagon



$$\text{Area of oct} \rightarrow 2a^2(1+\sqrt{2})$$

$$\text{Smaller diag} = a\sqrt{2+\sqrt{2}}$$

$$\text{Long diag} = \sqrt{2} \times (\text{smaller diag})$$

$$\text{Middle di} \leftarrow a(1+\sqrt{2})$$

Q17

Angle in a polygon

ABCDEFCA

$\angle BDF$

(we have to ~~reach~~ reach from B to F without crossing)
①

$$\text{Gap} = 4$$

$$\theta = \frac{360}{n} \times \text{Gap}$$

$$= \frac{360}{6} \times 4 \rightarrow 90^\circ$$

Sum of all angle in a Star Figure

$$(n-4)\pi$$

$$5\text{ star} \rightarrow \pi$$

$$6\text{ star} \rightarrow 2\pi$$

$$7\text{ star} \rightarrow 3\pi$$

$$8\text{ star} \rightarrow 4\pi$$

Q17

Angle in a polygon

ABCDEFCA

$\angle BDF$

(we have to ~~reach~~ reach from B to F without crossing)
①

$$\text{Gap} = 4$$

$$\theta = \frac{360}{n} \times \text{Gap}$$

$$= \frac{360}{6} \times 4 \rightarrow 90^\circ$$

Sum of all angle in a Star Figure

$$(n-4)\pi$$

$$5\text{ star} \rightarrow \pi$$

$$6\text{ star} \rightarrow 2\pi$$

$$7\text{ star} \rightarrow 3\pi$$

$$8\text{ star} \rightarrow 4\pi$$



Part 1d

Similar triangles

$$\frac{A_1}{A_2} = \frac{side_1^2}{side_2^2} = \frac{h_1^2}{h_2^2} = \frac{m_1^2}{m_2^2} = \frac{p_1^2}{p_2^2} = \frac{s_1^2}{s_2^2} = \frac{t_1^2}{t_2^2}$$

Part 1e

1 side

3, 4, 5
6, 8, 10
9, 12, 15
12, 16, 20
15, 20, 25
18, 24, 30

5, 12, 13
10, 24, 26
15, 36, 39

7, 24, 25

